

introduced a great deal of injustice. Our ability to reproduce and repeat performances allows me to listen on my laptop to hours of background music of the pianist Vladimir Horowitz (now extremely dead) performing Rachmaninoff's *Preludes*, instead of to the local Russian émigré musician (still living), who is now reduced to giving piano lessons to generally untalented children for close to minimum wage. Horowitz, though dead, is putting the poor man out of business. I would rather listen to Vladimir Horowitz or Arthur Rubinstein for \$10.99 a CD than pay \$9.99 for one by some unknown (but very talented) graduate of the Juilliard School or the Prague Conservatory. If you ask me why I select Horowitz, I will answer that it is because of the order, rhythm, or passion, when in fact there are probably a legion of people I have never heard about, and will never hear about—those who did not make it to the stage, but who might play just as well.

Some people naïvely believe that the process of unfairness started with the gramophone, according to the logic that I just presented. I disagree. I am convinced that the process started much, much earlier, with our DNA, which stores information about our selves and allows us to repeat our performance without our being there by spreading our genes down the generations. Evolution is *scalable*: the DNA that wins (whether by luck or survival advantage) will reproduce itself, like a bestselling book or a successful record, and become pervasive. Other DNA will vanish. Just consider the difference between us humans (excluding financial economists and businessmen) and other living beings on our planet.

Furthermore, I believe that the big transition in social life came not with the gramophone, but when someone had the great but unjust idea to invent the alphabet, thus allowing us to store information and reproduce it. It accelerated further when another inventor had the even more dangerous and iniquitous notion of starting a printing press, thus promoting texts across boundaries and triggering what ultimately grew into a winner-take-all ecology. Now, what was so unjust about the spread of books? The alphabet allowed stories and ideas to be replicated with high fidelity and without limit, without any additional expenditure of energy on the author's part for the subsequent performances. He didn't even have to be alive for them—death is often a good career move for an author. This implies that those who, for some reason, start getting some attention can quickly reach more minds than others and displace the competitors from the bookshelves. In the days of bards and troubadours, everyone had an audience. A storyteller, like a baker or a coppersmith, had a market, and the assurance that none from far away could dislodge him from his territory. Today, a few take almost everything; the rest, next to nothing.

By the same mechanism, the advent of the cinema displaced neighborhood